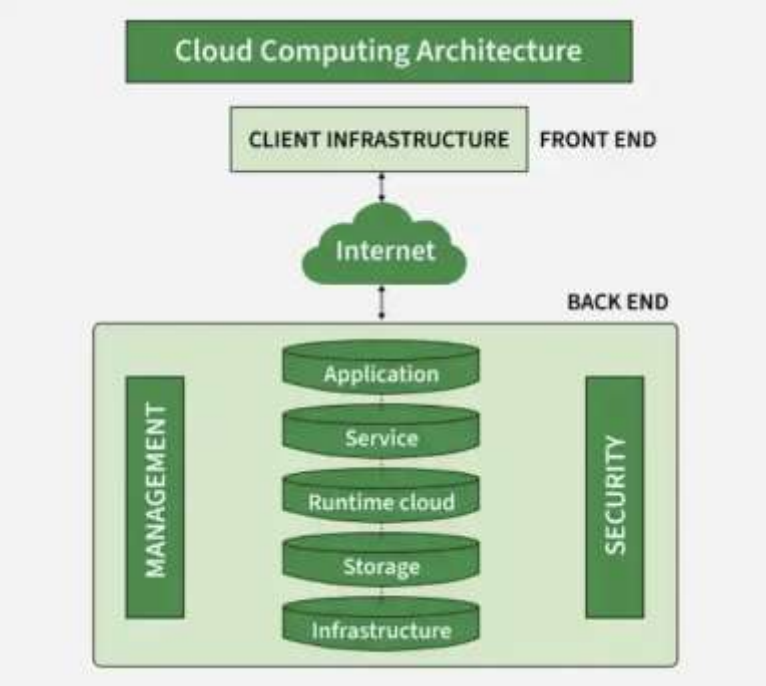
CLOUD COMPUTING BASICS **Date: 02-11-2026**

1. What is cloud computing?

Cloud computing is the on-demand delivery of IT resources and computing services such as servers, storage, databases, networking, and software over the internet with pay-as-you-go pricing.

- Companies had to purchase physical servers, storage devices, and networking equipment.
- Hardware is no longer purchased; computing resources are rented from cloud providers such as AWS, Google, or Microsoft.



Cloud Computing Architecture

2.What is Azure cloud?

Azure is a public cloud computing platform developed by Microsoft. It is used to build, test, deploy, and manage applications and services through Microsoft-managed data centers.

Key Things Azure is Used For:

App Hosting and Development	Data Storage and Databases
Artificial Intelligence and Machine Learning	Disaster Recovery and Backup
Hybrid and Multi-Cloud Operations	Security and Identity Management

Azure provides services across four main categories:

IaaS , PaaS , SaaS . Serverless Computing(Run code in response to events without provisioning or managing servers at all)

3. Advantages of cloud computing :

Key Advantages

- **Cost Savings & Pay-as-You-Go Model**: Replaces large upfront hardware capital expenditures (CapEx) with a flexible operational expenditure (OpEx) model where you pay only for the resources you consume.
- **Scalability and Flexibility:** Lets you scale compute power or storage up or down instantly to handle traffic spikes or quiet seasons without buying physical servers.
- **Global Accessibility and Mobility**: Enables teams to access files, applications, and data from anywhere in the world on any internet connected device.
- **Improved Collaboration**: Establishes a centralized "single source of truth" where multiple people can view and edit shared documents in real time, preventing version control conflicts.
- **Automatic Updates and Maintenance**: Offloads routine infrastructure upkeep, software patches, and hardware maintenance to the third-party provider.
- **Enhanced Security and Backup**: Provides enterprise grade features like automated backups, disaster recovery, data encryption, and remote wiping capabilities.
- **Faster Time to Market**: Allows developers to provision virtual instances and test new applications in minutes rather than waiting weeks for physical hardware procurement.

4. What is Iaas , PaaS , SaaS ?

1. IaaS (Infrastructure as a Service)

The most basic layer. It gives you raw, virtualized computing resources over the internet including virtual servers, storage space, and networking.

- **Your responsibility:** You manage the operating systems, middleware, applications, and data.
- **Provider's responsibility:** The vendor manages the physical hardware, data centers, and hypervisors.
- **Analogy:** Renting an empty building or plot of land where you build and set up everything yourself.
- **Examples:** Amazon Web Services (AWS) EC2, Google Compute Engine, and Azure Virtual Machines.
- **Target Audience**: IT Administrators, Network Architects, and DevOps teams.

2. PaaS (Platform as a Service)

A middle layer that provides a complete cloud-based development and deployment environment. It gives developers the tools, databases, and runtime systems needed to build software without worrying about the underlying servers.

- **Your responsibility:** You manage your custom code and application data.
- **Provider's responsibility:** The vendor manages the hardware, operating systems, and runtime software/tools.
- **Analogy:** Renting a fully equipped kitchen with ovens and counters you bring your own ingredients and cook your own meals.
- **Examples:** Google App Engine, Microsoft Azure App Services, and Heroku.
- **Target Audience:** Software Developers and Application Delivery Teams.

3. SaaS (Software as a Service)

A fully functional, ready-to-use software application delivered over the web through a browser or app. You just log in and use it.

- **Your responsibility:** You only manage your own user accounts and data inside the app.
- **Provider's responsibility:** The vendor manages everything else servers, storage, code, updates, security, and maintenance.
- **Analogy:** Going to a restaurant or buying a ready-made meal the chef handles everything, and you just eat.
- **Examples:** Google Gmail, Microsoft 365, Zoom, and Salesforce
- **Target Audience:** End-users, Business Professionals, and Consumers.

